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**B.Tech. Degree III Semester Regular/Supplementary Examination
February 2022**

**ME 19-205-0302 ELECTRICAL TECHNOLOGY
(2019 Scheme)**

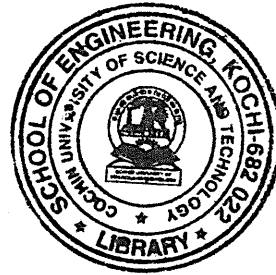
Time: 3 Hours

Maximum Marks: 60

**PART A
(Answer ALL questions)**

(8 × 3 = 24)

- I. (a) What are the precautions to be taken so that a DC generator not failed to build up emf?
- (b) Mention the different types of losses in DC machines.
- (c) Distinguish between potential transformer and current transformer.
- (d) Justify the statement "All power in an autotransformer is not transformed inductively".
- (e) Compare the two different types of rotors used in an alternator.
- (f) A 4-pole induction motor is energized from a 50 Hz supply system. If the machine runs on full load at 3% slip, determine the running speed and frequency of rotor currents.
- (g) Justify the statement "Load Dispatch Centre is known as Nerve Centre of power system".
- (h) Explain skin effect.



PART B

(4 × 12 = 48)

- II. (a) Explain the different speed control methods of DC shunt motor. (8)
 - (b) A DC shunt motor takes an armature current of 100 A at 480 V. The resistance of armature circuit is 0.2 Ω. The machine has 6 poles and armature is lap connected with 864 conductors. The flux per pole is 0.05 wb. Calculate (i) speed and (ii) torque. (4)
- OR**
- III. Explain the function of commutator in a DC machine. What are the effects of poor commutation on the performance of a DC machine? What methods are used to overcome these difficulties? (12)
- IV. (a) A 50 KVA single phase transformer draws a primary current of 250 A on full load. The total resistance referred to primary side is 0.006 Ω. If the iron loss of transformer is 200 W, calculate the efficiency on full load and on half load at 0.8 pf lagging. (8)
 - (b) Draw the phasor diagram of practical transformer at lagging power factor. (4)
- OR**
- V. A 5 kVA, 500/250 V, 50 Hz single phase transformer gave the following readings:- (12)
- OC test :- 500 V, 1 A, 50 W (LV side open)
- SC test:- 25 V, 10 A, 60 W (LV side shorted)
- (i) Determine the efficiency at 60% full load, 0.8 leading pf
 - (ii) Find load at which maximum efficiency occur
 - (iii) Determine the maximum efficiency of transformer
 - (iv) Draw the equivalent circuit referred to primary.

(P.T.O.)

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- VI. (a) A 1200 KVA, 50 Hz, three phase , star connected alternator has armature resistance of 0.25Ω per phase. A field current of 40 A produces a short circuit current of 200 A and an open circuit emf of 1100V line to line. Calculate the regulation on:
 (i) Full load 0.8 pf lagging
 (ii) Full load 0.8 pf leading. (8)
- (b) Explain different starting methods of synchronous motor. (4)
- OR**
- VII. (a) A 20 hp, 4 pole, three phase induction motor has friction and windage losses of 2% of output power. The full load slip is 3%. Calculate for full load
 (i) Rotor copper loss
 (ii) Rotor input
 (iii) Shaft torque. (8)
- (b) Derive the condition for maximum running torque in a three-phase induction motor. (4)
- VIII. (a) Discuss in detail the electrical equipment's used in power station. (8)
- (b) Explain the function of economizer and superheater in thermal power plant. (4)
- OR**
- IX. (a) With a neat schematic, explain the operation of a nuclear power plant. Mention its advantages and disadvantages. (8)
- (b) Define Corona. What are the factors affecting it? (4)
